

Linear Algebra for Political Scientists

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Structure

1. Matrices and Space
2. Finding Eigenvalues and Eigenvectors
3. Principal Component Analysis
4. Interpreting Principal Components
5. Wrap Up

What Does a Matrix Actually Do to Space?

- Session 1: Mechanical process of multiplying a matrix with a vector
- Session 2: Multiply matrix A with vector $\mathbf{x}$ in order to calculate composite scores
- Session 3: Multiply matrix X with vector $\hat{\beta}$ in order to calculate fitted values
- Session 4: What does multiplying by a matrix actually *do* to a vector?

In general, $A\mathbf{v}$ points in a different direction from $\mathbf{v}$ itself (and has a different length). So multiplying by a matrix usually *rotates* a vector and *rescales* it.

Visualizing Matrix-Vector Multiplication

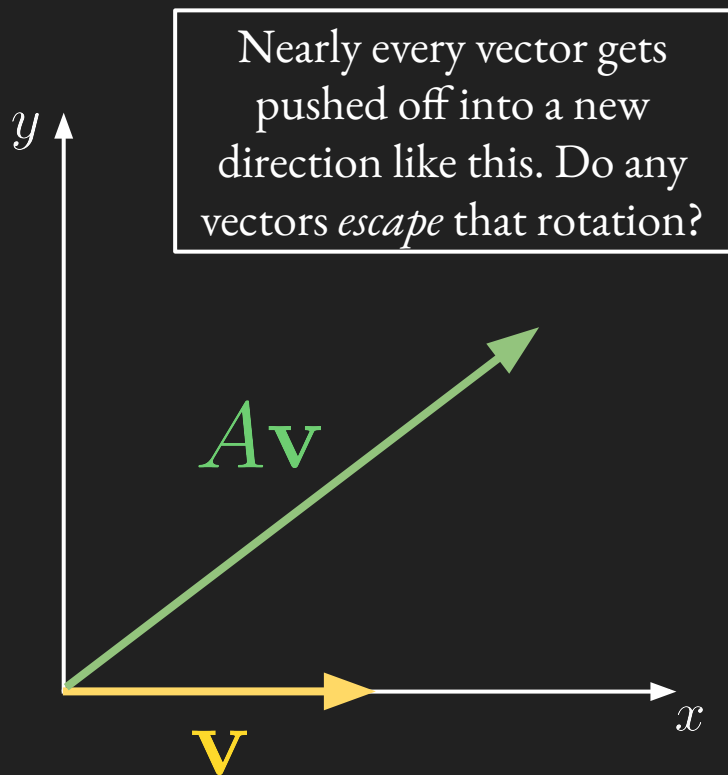
$$\mathbf{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$A\mathbf{v} = \begin{bmatrix} 2(1) + 1(0) \\ 1(1) + 2(0) \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Length changes, direction changes:

$$\|\mathbf{v}\| = \sqrt{1^2 + 0^2} = 1$$

$$\|A\mathbf{v}\| = \sqrt{2^2 + 1^2} = \sqrt{5}$$



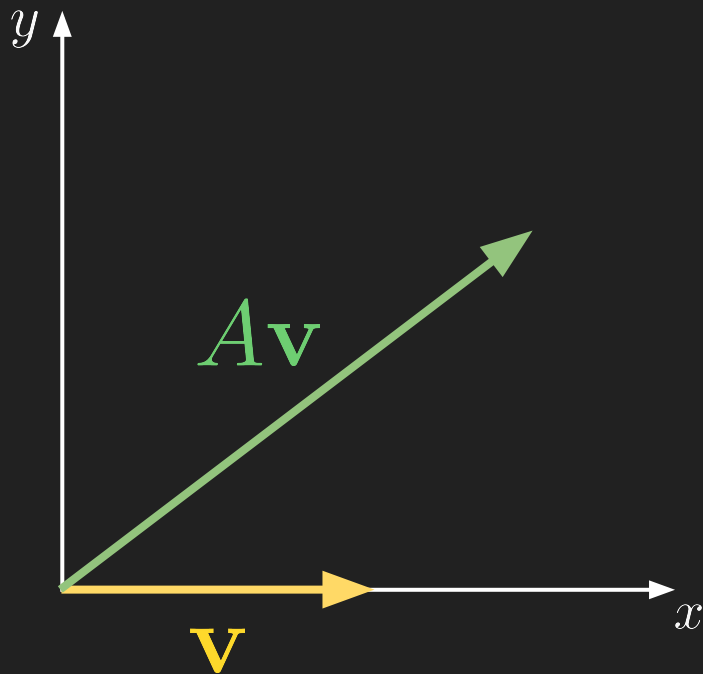
Eigenvectors and Eigenvalues

For a square matrix A , a nonzero vector $\mathbf{v}$ is an *eigenvector* of A if

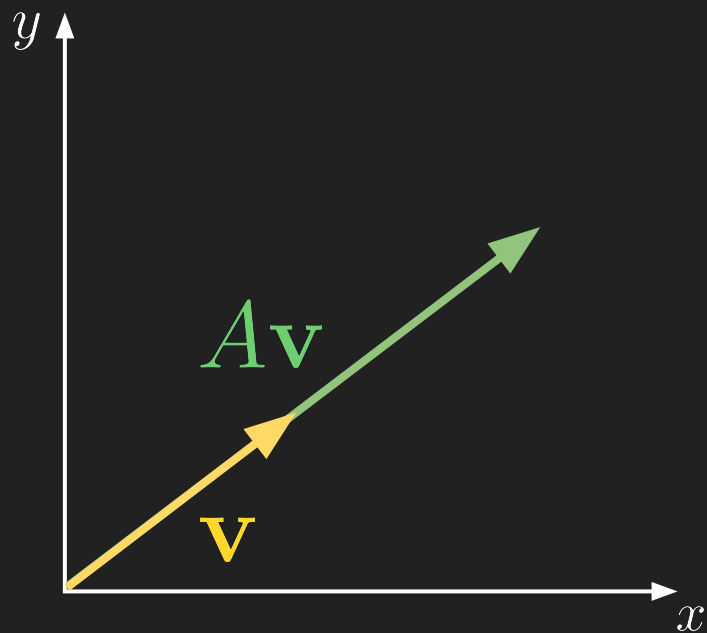
$$A\mathbf{v} = \lambda\mathbf{v}$$

for some scalar λ , called the corresponding *eigenvalue*. In words: A doesn't rotate $\mathbf{v}$ at all: it only stretches or shrinks it, by a factor of λ .

Most vectors are *not* eigenvectors of A . For a generic $\mathbf{v}$, $A\mathbf{v}$ will point somewhere new.



If $\mathbf{v}$ is an eigenvector of A , multiplying A by $\mathbf{v}$ simply *scales* $\mathbf{v}$ in the same direction.



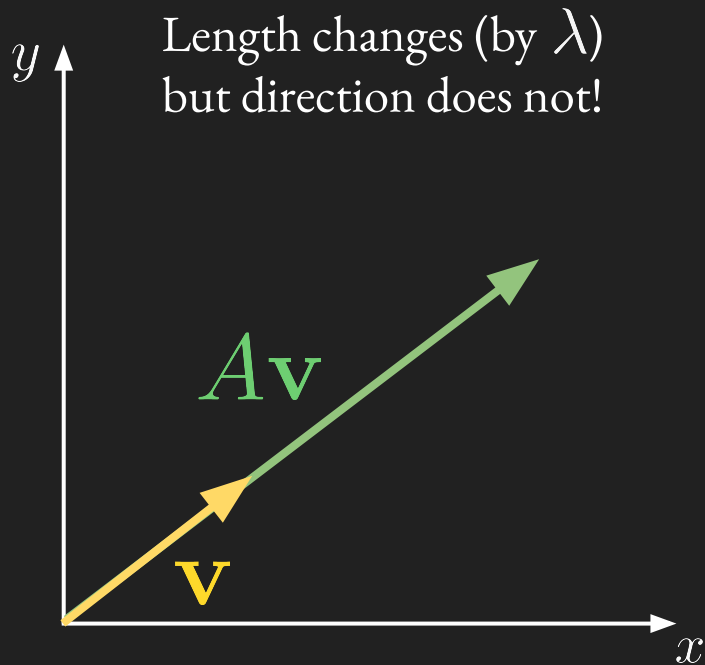
Matrix-Eigenvector Multiplication

$$\mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$A\mathbf{v} = \begin{bmatrix} 2(1) + 1(1) \\ 1(1) + 2(1) \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} = \overset{\lambda}{\textcircled{3}}\mathbf{v}$$

$$\|\mathbf{v}\| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\|A\mathbf{v}\| = \sqrt{3^2 + 3^2} = 3\sqrt{2}$$



Finding Eigenvectors & Eigenvalues Systematically

$$\begin{aligned} A\mathbf{v} &= \lambda\mathbf{v} \Leftrightarrow A\mathbf{v} - \lambda\mathbf{v} = \mathbf{0} \\ (A - \lambda I)\mathbf{v} &= \mathbf{0} \end{aligned}$$

The only way for some nonzero $\mathbf{v}$ to satisfy the above equation is for $A - \lambda I$ to be non-invertible. So, the eigenvalues of A are the exactly the solution to:

$$\boxed{\det(A - \lambda I) = 0}$$

Characteristic Equation

Finding Eigenvalues: Example

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{bmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{vmatrix} = (2 - \lambda)(2 - \lambda) - (1)(1) = 0$$

$$(\lambda - 1)(\lambda - 3) = 0 \quad \implies \quad \lambda_1 = 1, \quad \lambda_2 = 3$$

Finding Eigenvectors: Example

$\lambda_1 = 1 \longrightarrow$ Plug back into $(A - \lambda I)\mathbf{v} = \mathbf{0}$:

$$\begin{bmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \longrightarrow \begin{array}{l} v_1 + v_2 = 0 \\ v_1 = -v_2 \end{array} \longrightarrow \mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$\lambda_2 = 3 \longrightarrow$ Plug back into $(A - \lambda I)\mathbf{v} = \mathbf{0}$:

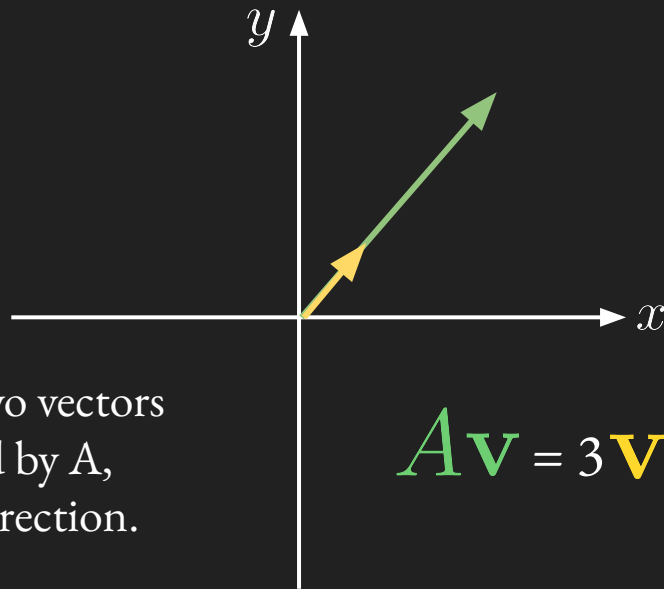
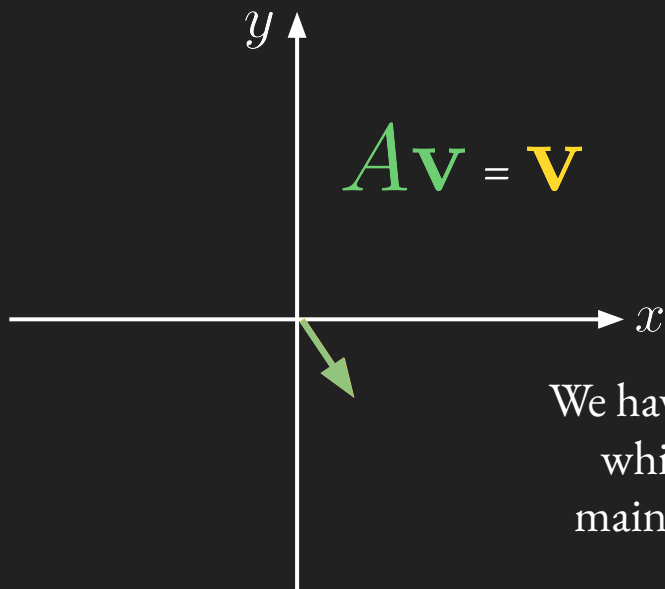
$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \longrightarrow \begin{array}{l} -v_1 + v_2 = 0 \\ v_1 = v_2 \end{array} \longrightarrow \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$A\mathbf{v}_1 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$A\mathbf{v}_2 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$



We have found the only two vectors
which, when multiplied by A ,
maintain their original direction.

The Spectral Theorem

Notice that the two eigenvectors we just found are orthogonal to each other:

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1 - 1 = 0$$

This is a general property of symmetric matrices resulting from the Spectral Theorem:

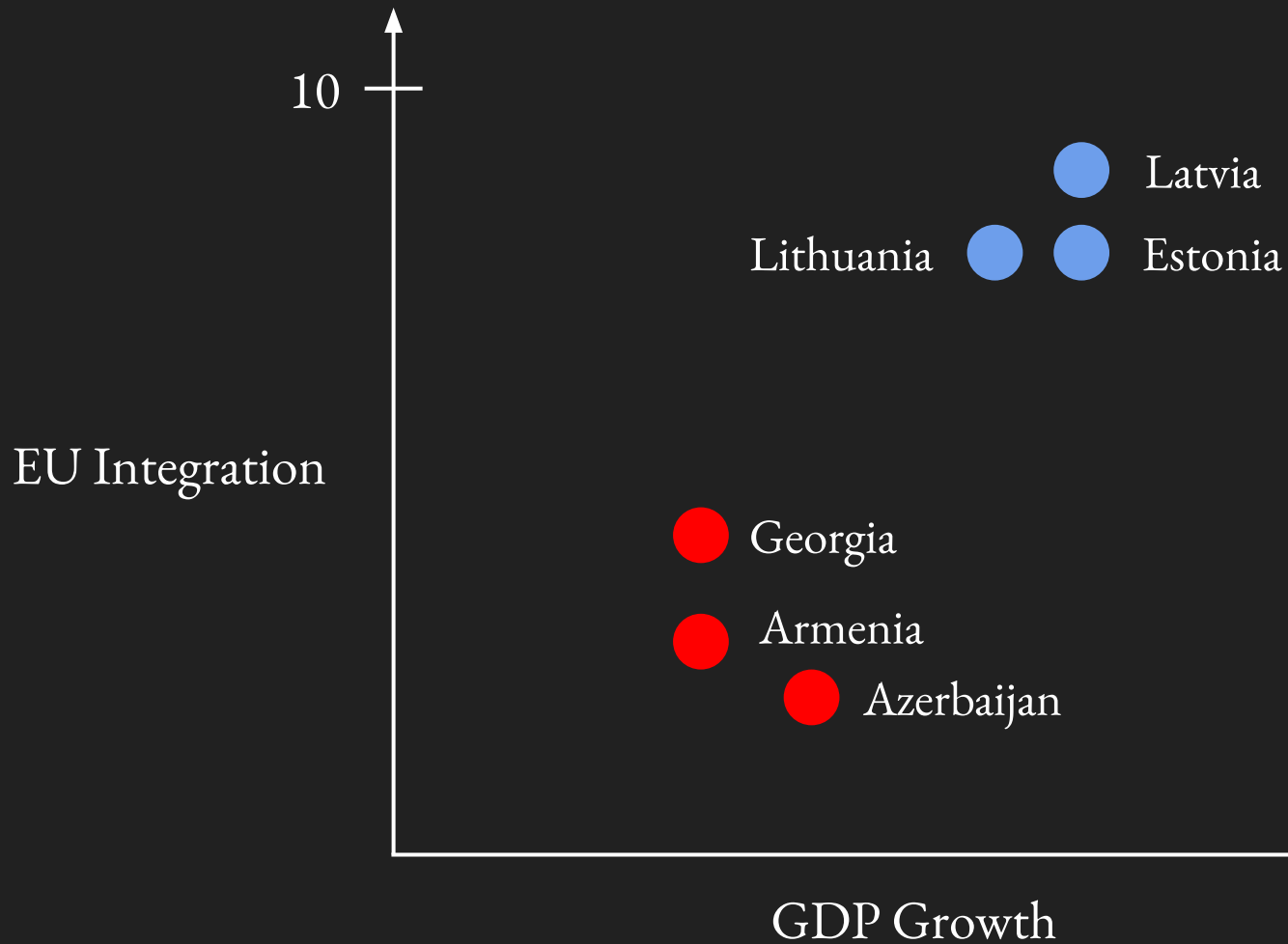
If S is a symmetric matrix, then:

1. All of S 's eigenvalues are real numbers (never complex).
2. Eigenvectors corresponding to different eigenvalues are orthogonal to each other.

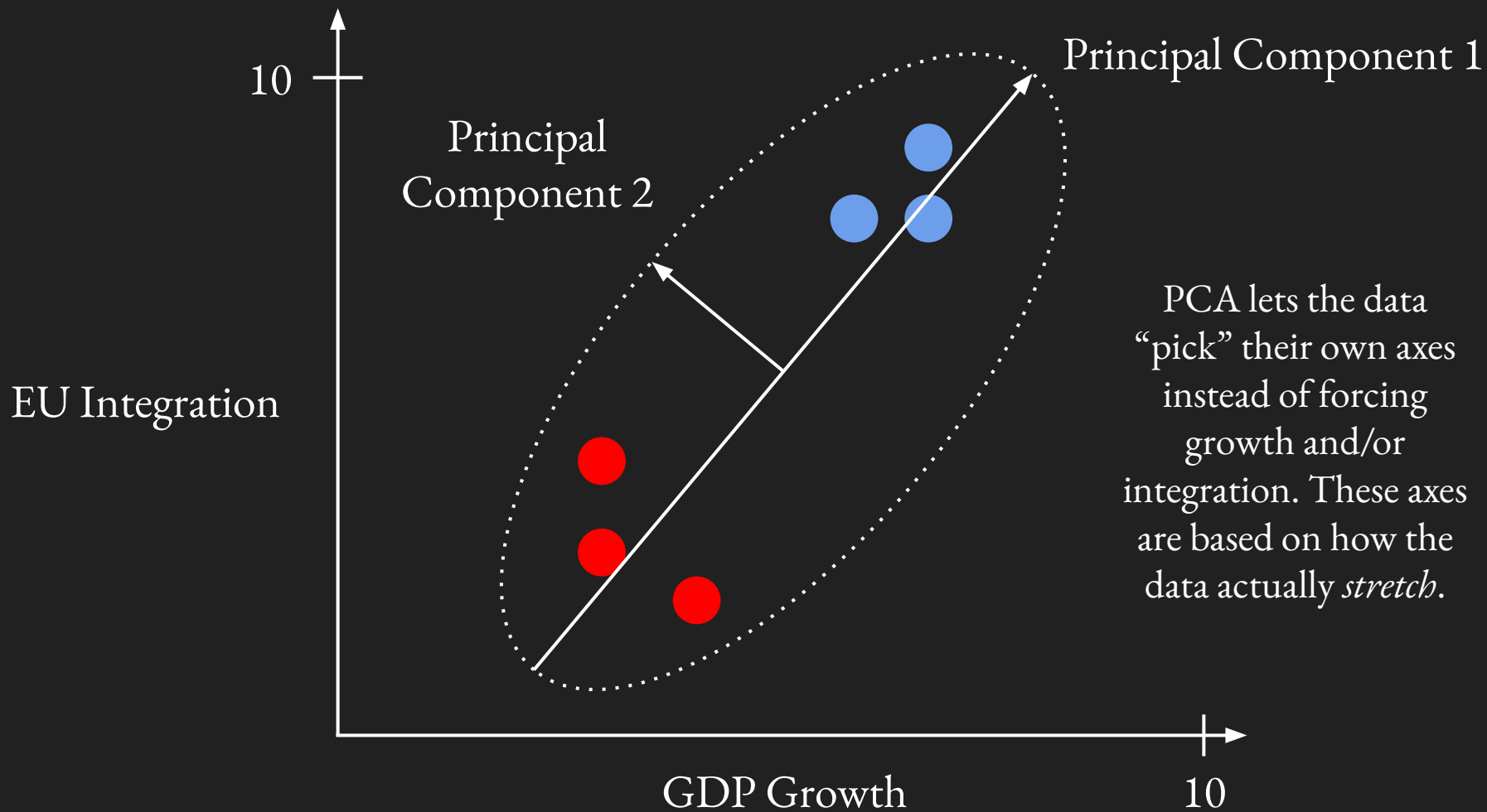
Principal Component Analysis: Motivation

- The Problem: Political scientists constantly end up with many **correlated measurements** of the same underlying concept (e.g., a battery of survey items, a dozen V-Dem democracy indicators, economics indices, etc.).
- PCA gives us a way to extract the **independent** information embedded in those correlated columns in just a couple of new variables (components).
- Why not just create another index, like we did in Session 2?
 - Not all columns are independent, so we may inflate the index with redundant information.
 - Summing the columns will reduce the information to a scalar. We are interested in what inherent dimensionality the data “inhabits.”

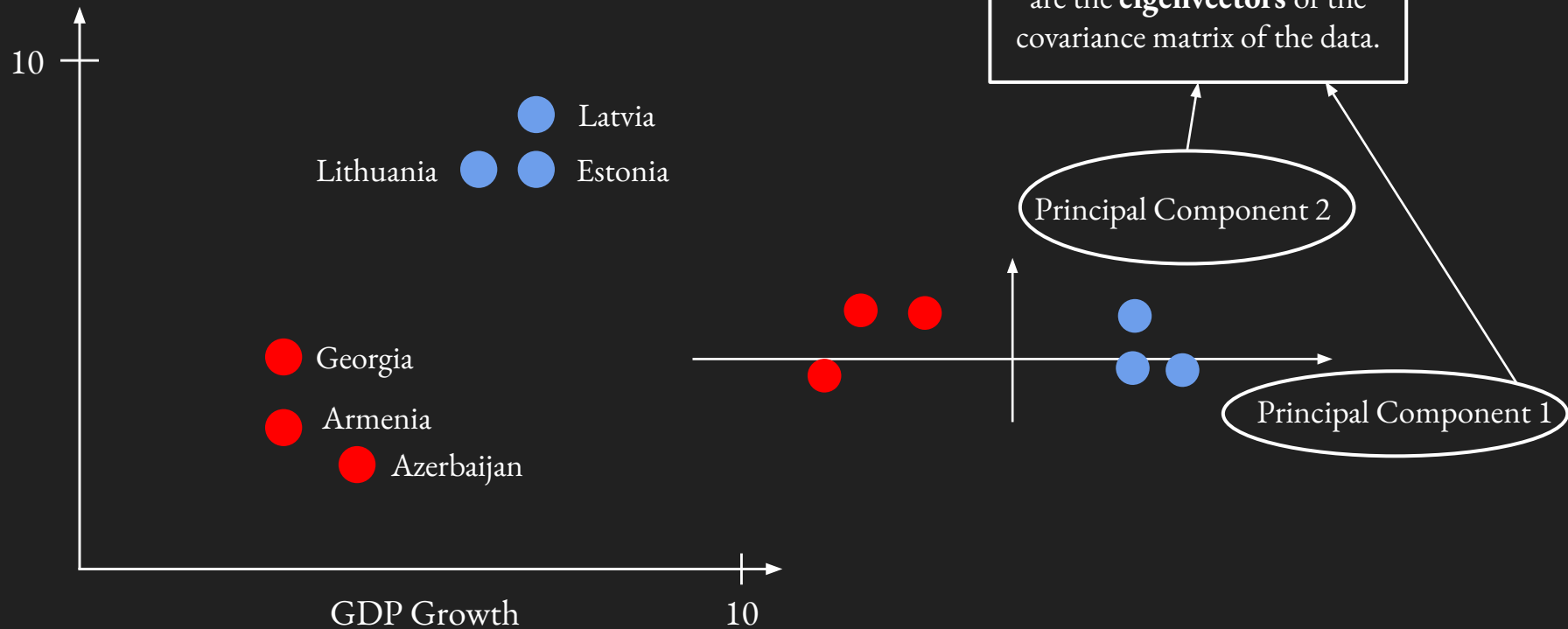
Country	Region	GDP Growth	EU Integration	Democracy Score
Armenia	South Caucasus	3	2	3
Azerbaijan	South Caucasus	4	1	2
Georgia	South Caucasus	3	3	4
Estonia	Baltics	7	9	8
Latvia	Baltics	6	8	7
Lithuania	Baltics	7	8	8



Along which **single** axis do these countries differ the most?



EU Integration



PCA doesn't change the data; rather, it finds the rotation of the axes that captures the most amount of variation on the first axis (Principal Component 1).

Deriving Principal Components

$$X = \begin{bmatrix} 3 & 2 \\ 4 & 1 \\ 3 & 3 \\ 7 & 9 \\ 6 & 8 \\ 7 & 8 \end{bmatrix}$$

GDP EU
Growth Integration

Covariance Matrix of X

$$\Sigma = \begin{bmatrix} \text{Var}(\text{growth}) & \text{Cov}(\text{growth, integration}) \\ \text{Cov}(\text{integration, growth}) & \text{Var}(\text{integration}) \end{bmatrix}$$
$$= \begin{bmatrix} \frac{18}{5} & \frac{31}{5} \\ \frac{31}{5} & \frac{377}{30} \end{bmatrix} \rightarrow$$

The eigenvectors of this matrix are the principal components!

And since this matrix is symmetric, the Spectral Theorem guarantees that the eigenvectors will be **orthogonal** and the eigenvalues will be **real**.

Calculating Principal Components as Eigenvectors

$$\Sigma = \begin{bmatrix} \frac{18}{5} & \frac{31}{5} \\ \frac{31}{5} & \frac{377}{30} \end{bmatrix}$$

$$\begin{vmatrix} \frac{18}{5} - \lambda & \frac{31}{5} \\ \frac{31}{5} & \frac{377}{30} - \lambda \end{vmatrix} = 30\lambda^2 - 485\lambda + 204 = 0$$

$$\lambda_1 \approx 15.73, \lambda_2 \approx 0.43$$

Plugging back into $(\Sigma - \lambda I)\mathbf{v} = \mathbf{0}$ and normalizing gives us $\mathbf{v}_1 \approx \begin{bmatrix} 0.455 \\ 0.890 \end{bmatrix}$

Since integration's weight is nearly double that of growth, the direction our six countries differ **most** along leans on integration more than growth.

$$\mathbf{v}_1 \approx \begin{bmatrix} 0.455 \\ 0.890 \end{bmatrix}$$

The eigenvector of Σ .

The long axis of the data's ellipse.

A new variable: Since this vector's entries are weights, it also defines a new variable outright: $\text{PC1} = 0.455(\text{growth}) + 0.890(\text{integration})$, applied to any country's (centered) data.

The single direction, out of infinitely many, along which projecting the data explains as much of its spread as mathematically possible.

Reducing Many Columns to a Few Dimensions

The real payoff of PCA shows up once there are many variables. Consider a battery of questions from the World Values Survey:

To what extent do you trust people in your family?
To what extent do you trust people in your neighborhood?
To what extent do you trust people you know personally?
To what extent do you trust people you meet for the first time?
To what extent do you trust people of another religion?
To what extent do you trust people of another nationality?

These columns
are likely **highly**
correlated!

Idea: With 6 correlated columns (and ~thousands of rows), keeping every answer is a lot of raw data to interpret. We want a more compact summary of how many “different” directions of variation there are.

PCA offers us exactly this.

Someone who reports high trust in strangers of another nationality and/or religion is very likely to also report relatively high trust in people they meet for the first time in general.



Respondent	Family	Neighborhood	Know Personally	Meet First Time	Religion	Nationality
1						
2						
⋮						

PCA: General Procedure

0. Center each variable.
1. Compute the covariance matrix, Σ .
2. Solve the characteristic equation $\det(\Sigma - \lambda I) = 0$ to obtain the eigenvalues.
3. Calculate the eigenvector associated with each eigenvalue.
4. Order and label eigenvectors (eigenvector associated with largest eigenvalue is PC1, etc.).
5. Check how much variance each PC captures.*

*Upcoming!

$$\Sigma - \lambda I = \begin{bmatrix} 1 - \lambda & \rho_{12} & \rho_{13} & \rho_{14} & \rho_{15} & \rho_{16} \\ \rho_{12} & 1 - \lambda & \rho_{23} & \rho_{24} & \rho_{25} & \rho_{26} \\ \rho_{13} & \rho_{23} & 1 - \lambda & \rho_{34} & \rho_{35} & \rho_{36} \\ \rho_{14} & \rho_{24} & \rho_{34} & 1 - \lambda & \rho_{45} & \rho_{46} \\ \rho_{15} & \rho_{25} & \rho_{35} & \rho_{45} & 1 - \lambda & \rho_{56} \\ \rho_{16} & \rho_{26} & \rho_{36} & \rho_{46} & \rho_{56} & 1 - \lambda \end{bmatrix}$$

Beyond 3x3 matrices, the calculation of eigenvalues becomes unwieldy. So we will rely on statistical software to calculate them for us.

Total Variance Explained

$$\lambda_1 \approx 3.2, \lambda_2 \approx 1.3, \lambda_3 \approx 0.5, \lambda_4 \approx 0.4, \lambda_5 \approx 0.3, \lambda_6 \approx 0.3$$

$$\text{Variance explained by PC}_i = \frac{\lambda_i}{\sum_{j=1}^k \lambda_j}$$

So, the first principal component explains $\frac{3.2}{6} \approx 53\%$ of the total variance.

$\lambda_1 + \lambda_2$ together explain about 75% of the total variance.

Two components (not six!) capture most of what these questions are telling us.

Principal Component Loadings

Item	Principal Component 1	Principal Component 2
Family	0.30	0.55
Neighborhood	0.35	0.50
Know Personally	0.40	0.10
First Time	0.45	-0.15
Religion	0.45	-0.35
Nationality	0.45	-0.40

All of these weights are called **loadings**. They tell us the weight the original variable carries in that PC.

Sign: A positive loading means that variable moves in the same direction as the component.

Magnitude: Bigger (in absolute value) means that variable contributes more to the component.

When several variables all have large, same-signed loadings on one component, this is a signal they're all tapping into the same underlying pattern.

Interpreting Principal Components

Item	Principal Component 1
Family	0.30
Neighborhood	0.35
Know Personally	0.40
First Time	0.45
Religion	0.45
Nationality	0.45

We could call Principal Component 1
“generalized trust.”

All six loadings are positive and similarly sized, so this axis just tracks how trusting a respondent is overall: someone high on PC1 trusts family, neighbors, strangers, and out-groups all somewhat more than average; someone low on PC1 trusts all of them somewhat less.

Interpreting Principal Components (II)

Item	Principal Component 2
Family	0.55
Neighborhood	0.50
Know Personally	0.10
First Time	-0.15
Religion	-0.35
Nationality	-0.40

We could call Principal Component 2 “particularized vs. generalized trust.”

Family and neighborhood load positively, the three out-group items load negatively. This axis picks up respondents who trust their close, familiar circle but not outsiders (high PC2) versus respondents whose trust extends more evenly to strangers and out-groups (low/negative PC2).

Words of Caution: Interpreting Principal Components

- Sign is relative: $-\mathbf{v}$ is just as valid an eigenvector as $\mathbf{v}$, so whether “high” on a component means one thing or its opposite is a labeling choice.
- A name for a component is your interpretation: it is up to **you** to figure out the substantive meaning of your components.
- A variable with bigger raw units will mechanically dominate the loadings, so run PCA on standardized variables.
- Components are orthogonal by construction, but this doesn't necessarily mean that that they will represent substantively independent ideas.

Session 1

Session 2

Session 3

Session 4

Vectors, Matrices

Systems of Equations

Column Space

Eigenvalues

Vector Operations

Gaussian Elimination

Orthogonal
Projection

Eigenvectors

Dot Product

Column Space

Normal Equations

Characteristic
Equation

Orthogonality

Row Echelon Form

OLS Estimator

Spectral Theorem

Matrix
Multiplication

Matrix Rank

Covariance Matrices

Principal
Component
Analysis

Inverse

Special Matrices

Standard Errors

Interpreting PCs

Determinant

Collinearity